

Sr. No	Question Content	Option 1	Option 2	Option 3	Option 4	Correct
1	Software is defined as _____	Set of programs, documentation & configuration of data	Set of programs	documentation and configuration of data	None of the mentioned	Set of programs, documentation & configuration of data
2	What is Software Engineering?	Designing a software	Testing a software	Application of engineering principles to the design a software	None of the above	Application of engineering principles to the design a software
3	Who is the father of Software Engineering?	Margaret Hamilton	Watts S. Humphrey	Alan Turing	Boris Beizer	Watts S. Humphrey
4	_____ is a software development activity that is not a part of software processes.	Validation	Specification	Development	Dependence	Dependence
5	Which of the following is NOT a characteristic of software?	Software is developed or engineered	Software does not wear out	Software is manufactured	Software is easily destroyed	Software is easily destroyed
6	In the software engineering context, which of the following describes a software component?	A part of the software that can be reused	A complete application	A programming language	A set of hardware specifications	A part of the software that can be reused
7	Which of the following is NOT an application area of software engineering?	Web development	Database management	Music production	Automotive engineering	Music production
8	What is the primary goal of layered technologies in software engineering?	To increase software cost	To enhance system performance and security	To abstract complexity	To reduce code size	To abstract complexity
9	Which process model is characterized by a linear sequential flow?	Incremental model	Agile model	Waterfall model	Spiral model	Waterfall model
10	In which model is software developed in small, incremental releases?	Waterfall model	Incremental model	Spiral model	Prototyping model	Incremental model
11	The Prototyping Model is particularly useful when:	Requirements are well-understood	Requirements are unclear or evolving	Time constraints are strict	The project is large	Requirements are unclear or evolving
12	What is a key advantage of the Spiral Model?	It is the least expensive	It is a purely sequential model	It allows for risk assessment at each iteration	It requires less documentation	It allows for risk assessment at each iteration
13	In Agile development, which principle emphasizes customer collaboration?	Following a plan	Working software	Individuals and interactions	Customer collaboration over contract negotiation	Customer collaboration over contract negotiation
14	What does the term 'process' refer to in software engineering?	A programming language	A sequence of tasks performed to develop software	A set of documentation practices	A testing methodology	A sequence of tasks performed to develop software
15	Which of the following is NOT a layer in the layered technology model?	Tool	Method	Process	Presentation	Presentation
16	Which process model allows for concurrent development of different components?	Waterfall model	Incremental model	Spiral model	Agile model	Agile model
17	What is the primary focus of the Waterfall Model?	Flexibility in requirements	A sequential phase completion	Early user feedback	Risk management	A sequential phase completion

18	Which of the following tools is commonly used for software development?	Word processor	Version control systems	Spreadsheet software	Presentation software	Version control systems
19	The Agile Manifesto values:	Processes and tools over individuals	Comprehensive documentation over working software	Customer collaboration over contract negotiation	Following a plan over responding to change	Customer collaboration over contract negotiation
20	Which of the following is a disadvantage of the Waterfall Model?	Clear documentation	High risk and uncertainty	Simple to manage	Easy to understand	High risk and uncertainty
21	What is the main purpose of software process models?	To eliminate the need for documentation	To provide a framework for planning, structuring, and controlling the process	To restrict flexibility in software development	To enhance hardware specifications	To provide a framework for planning, structuring, and controlling the process
22	Which model emphasizes iterative development and risk assessment?	Waterfall model	Incremental model	Spiral model	Prototyping model	Spiral model
23	Which characteristic best describes software maintenance?	It involves fixing defects only	It is unnecessary for software	It is the process of modifying a software system after delivery	It is the same as software development	It is the process of modifying a software system after delivery
24	The Incremental Model is best used when:	All requirements are known in advance	The system needs to be built quickly	The software is complex and needs to evolve	It is unnecessary to release partial systems	The system needs to be built quickly
25	In which model do risk assessment and mitigation play a crucial role?	Agile model	Waterfall model	Spiral model	Incremental model	Spiral model
26	Which of the following statements about the Agile model is true?	It requires detailed documentation at every step	It discourages customer involvement	It promotes adaptive planning and encourages rapid responses to change	It is exclusively suitable for small projects	It promotes adaptive planning and encourages rapid responses to change
27	What is the primary purpose of using software engineering tools?	To replace human effort	To enhance productivity and facilitate software development	To make software more complex	To ensure compliance with regulations	To enhance productivity and facilitate software development
28	In the context of software engineering, what is 'refactoring'?	Adding new features without changing the existing behavior	Redesigning the architecture	Fixing bugs in the software	Improving code structure without changing its functionality	Improving code structure without changing its functionality
29	Which of the following models allows for iterative development?	Waterfall model	Incremental model	Spiral model	Both B and C	Both B and C
30	Which of the following is NOT a common Agile framework?	Scrum	Kanban	Waterfall	Extreme Programming (XP)	Waterfall
31	What is a significant advantage of using the Spiral Model?	Low cost	It eliminates the need for documentation	Continuous user feedback through iterations	Strictly follows a sequential approach	Continuous user feedback through iterations
32	In software development, what does 'requirements gathering' involve?	Writing code for the software	Understanding and documenting user needs and expectations	Testing the software	Deploying the software	Understanding and documenting user needs and expectations
33	Which of the following is a primary role of a software engineer?	Market analysis	Designing and developing software systems	Customer support	Hardware maintenance	Designing and developing software systems
34	What does the term 'continuous integration' refer to in Agile development?	Keeping software features isolated	Regularly merging code changes into a central repository	Completing a project in one go	Documenting every code change	Regularly merging code changes into a central repository
35	Which model is best suited for projects where requirements are well-defined and unlikely to change?	Spiral model	Waterfall model	Agile model	Prototyping model	Waterfall model

36	What is the primary purpose of a project scope statement?	To outline the project budget	To define what is included and excluded in the project	To assign project team roles	To identify project stakeholders	To define what is included and excluded in the project
37	Feasibility studies are typically conducted during which phase of a project?	Initiation	Planning	Execution	Closure	Initiation
38	Which of the following is NOT a component of a feasibility study?	Technical feasibility	Economic feasibility	Environmental feasibility	Marketing feasibility	Marketing feasibility
39	What technique uses historical data to estimate future project tasks?	Analogous estimation	Parametric estimation	Bottom-up estimation	Three-point estimation	Analogous estimation
40	Which of the following is a common method for estimating project costs?	Monte Carlo analysis	Critical path analysis	Delphi technique	Resource leveling	Critical path analysis
41	In effort estimation, what does the term “three-point estimate” refer to?	Using three different teams to estimate effort	Estimating best case, worst case, and most likely scenarios	A method involving three project phases	A standard deviation calculation	Estimating best case, worst case, and most likely scenarios
42	Which of the following is a disadvantage of expert judgment in effort estimation?	Time-consuming	Can be biased	Inaccurate	Costly	Can be biased
43	What is the primary goal of project scheduling?	To control project costs	To allocate resources efficiently	To determine project timelines	To assess project quality	To determine project timelines
44	In project management, what does the term 'resource leveling' refer to?	Distributing project resources evenly	Adjusting project schedules to balance resource demands	Evaluating team member performance	Prioritizing project tasks	Adjusting project schedules to balance resource demands
45	What is a common approach for developing a project schedule?	Agile methodology	Waterfall methodology	Critical path method	Lean methodology	Critical path method
46	What is the main focus of quality planning in project management?	Identifying project risks	Defining quality standards and metrics	Assigning team roles	Estimating project costs	Defining quality standards and metrics
47	Which of the following is a tool used in quality planning?	Gantt chart	RACI matrix	Control chart	SWOT analysis	Control chart
48	What does the term “quality assurance” refer to?	Measuring project deliverables	Activities to ensure quality standards are met	Identifying quality risks	Developing quality metrics	Activities to ensure quality standards are met
49	Which standard is commonly used for quality management in projects?	ISO 9001	PMBOK	Six Sigma	PRINCE2	ISO 9001
50	What is the role of the quality management plan?	To define how quality will be managed throughout the project	To identify project stakeholders	To estimate project costs	To allocate resources	To define how quality will be managed throughout the project
51	What is the first step in the risk management process?	Risk control	Risk identification	Risk assessment	Risk monitoring	Risk identification
52	What tool is commonly used to assess the probability and impact of risks?	Risk matrix	Gantt chart	SWOT analysis	Fishbone diagram	Risk matrix
53	What is a risk response strategy that involves taking steps to reduce the likelihood of a risk occurring?	Acceptance	Mitigation	Avoidance	Transference	Mitigation

54	Which of the following describes “risk acceptance”?	Avoiding the risk altogether	Transferring the risk to another party	Acknowledging the risk and choosing to proceed	Implementing measures to reduce risk	Acknowledging the risk and choosing to proceed
55	What is the purpose of a risk register?	To document project deliverables	To track identified risks and their management	To outline project scope	To monitor project costs	To track identified risks and their management
56	What is the primary purpose of project monitoring and controlling?	To define project scope	To ensure project objectives are met	To allocate resources	To develop project schedules	To ensure project objectives are met
57	Which of the following metrics is commonly used to track project performance?	Earned Value Management (EVM)	Cost-benefit analysis	SWOT analysis	Fishbone diagram	Earned Value Management (EVM)
58	What does the term “baseline” refer to in project management?	The final project report	The approved version of a project plan	The project scope	The project schedule	The approved version of a project plan
59	Which tool can be used for tracking project progress over time?	Resource histogram	Control chart	Pareto chart	Gantt chart	Gantt chart
60	In project monitoring, what is the purpose of variance analysis?	To identify new risks	To compare planned vs. actual performance	To assess stakeholder engagement	To develop project budgets	To compare planned vs. actual performance
61	What is the primary focus of detailed scheduling?	Identifying project risks	Allocating resources effectively	Planning individual tasks and their timelines	Assessing project quality	Planning individual tasks and their timelines
62	Which of the following is NOT typically included in a detailed project schedule?	Task durations	Resource assignments	Risk management plan	Task dependencies	Risk management plan
63	What scheduling technique is used to determine the longest path through a project?	Gantt chart	PERT analysis	Critical Path Method (CPM)	Resource leveling	Critical Path Method (CPM)
64	What is the first step in the requirement engineering process?	Requirement specification	Problem recognition	Requirement validation	Requirement analysis	Problem recognition
65	Which of the following is a common technique used for problem recognition?	Surveys	Interviews	SWOT analysis	Root cause analysis	Root cause analysis
66	In requirement engineering, what is the main purpose of requirement elicitation?	To document requirements	To identify stakeholders	To gather requirements from stakeholders	To validate requirements	To gather requirements from stakeholders
67	Which of the following is NOT a requirement engineering task?	Requirement elicitation	Requirement validation	Requirement testing	Requirement specification	Requirement testing
68	What is the goal of requirement specification?	To gather user feedback	To document detailed requirements	To analyze risks	To define project scope	To document detailed requirements
69	What type of requirement specifies what the system should do?	Non-functional requirement	Functional requirement	Performance requirement	Technical requirement	Functional requirement
70	Which document typically contains functional requirements?	Requirement specification document	Design document	Test plan	Project charter	Requirement specification document
71	Use cases are primarily used to capture what aspect of requirements?	Non-functional requirements	Functional requirements	Design requirements	Implementation details	Functional requirements

72	In a use case diagram, what does an actor represent?	A user or system that interacts with the system	A specific use case	A requirement type	A functional component	A user or system that interacts with the system
73	What is a primary benefit of using use cases?	They provide detailed design specifications	They help to identify system interactions	They eliminate the need for testing	They focus solely on performance metrics	They help to identify system interactions
74	What is the purpose of requirements validation?	To gather requirements from stakeholders	To ensure requirements are correct and complete	To analyze the requirements	To prioritize requirements	To ensure requirements are correct and complete
75	Which of the following techniques is commonly used for requirements validation?	Prototyping	Code reviews	Performance testing	Static analysis	Prototyping
76	What is the main focus of requirement analysis?	To document requirements	To assess the feasibility of requirements	To create test cases	To design the system architecture	To assess the feasibility of requirements
77	Which of the following is a characteristic of good requirements?	Ambiguity	Vague descriptions	Measurable	Subjectivity	Measurable
78	Which technique involves creating a model of the system to validate requirements?	Use case modeling	Process modeling	Prototyping	Context diagramming	Prototyping
79	In requirement engineering, what does the term "stakeholder" refer to?	A user of the system	Anyone who has an interest in the project	A project manager	A team member	Anyone who has an interest in the project
80	Which of the following best describes a non-functional requirement?	It specifies a particular behavior of the system	It defines how the system should perform under certain conditions	It describes a specific feature of the system	It is irrelevant to system behavior	It defines how the system should perform under certain conditions
81	What is a common pitfall in requirement specification?	Including too many details	Focusing too much on non-functional requirements	Using technical jargon	Not involving stakeholders	Not involving stakeholders
82	Which document is used to track changes in requirements throughout the project lifecycle?	Requirement specification document	Change request document	Traceability matrix	Project plan	Traceability matrix
83	What does traceability in requirements engineering refer to?	The ability to track changes in requirements	The relationship between requirements and their implementation	The accuracy of requirement specifications	The validation of requirements	The relationship between requirements and their implementation
84	What is a key outcome of the requirement elicitation process?	Identifying potential risks	Generating a requirements list	Developing a project plan	Creating design specifications	Generating a requirements list
85	In a use case, what is the "main success scenario"?	The ideal flow of events when everything goes right	The alternative flows when something goes wrong	The overall system performance	The user interface design	The ideal flow of events when everything goes right
86	Which of the following is a method for gathering requirements from users?	Code review	Prototyping	Interviews	Debugging	Interviews
87	What is the primary purpose of a requirement specification document?	To outline the project timeline	To communicate requirements to stakeholders	To track project costs	To manage project risks	To communicate requirements to stakeholders
88	What is the role of prototypes in requirement validation?	They replace the need for documentation	They help users visualize and validate requirements	They are only used for testing	They are not useful in requirement validation	They help users visualize and validate requirements
89	What type of requirement would specify response time for a web application?	Functional requirement	Non-functional requirement	Technical requirement	User requirement	Non-functional requirement

90	Which of the following is essential for ensuring requirements are understood by all stakeholders?	Technical jargon	User training	Clear and concise language	Strict adherence to standards	Clear and concise language
91	What is a common reason for requirement changes during a project?	Market conditions	Team member changes	Budget constraints	All of the above	All of the above
92	How can requirements be prioritized effectively?	By stakeholder voting	By assigning numerical values based on importance	By random selection	By team discussions only	By assigning numerical values based on importance
93	What does SWOT stand for?	Strengths, Weaknesses, Opportunities, Threats	Systems, Workflows, Objectives, Tactics	Strengths, Weaknesses, Options, Trends	Systems, Weaknesses, Objectives, Tactics	Strengths, Weaknesses, Opportunities, Threats
94	What is the primary purpose of a Gantt chart?	To list all project resources	To visually represent a project schedule	To calculate project costs	To define project scope	To visually represent a project schedule
95	What is a key advantage of using a Gantt chart in project management?	It eliminates the need for scheduling	It provides a clear visual overview of project timelines	It offers detailed cost breakdowns	It focuses solely on team roles	It provides a clear visual overview of project timelines
96	Which software tool is commonly used to create Gantt charts?	Microsoft Word	Microsoft Excel	Microsoft PowerPoint	Microsoft Project	Microsoft Project
97	In project management, which aspect does a Gantt chart NOT typically address?	Task timelines	Resource allocation	Budget estimation	Risk management	Risk management
98	Which one of the following activities is not recommended for software processes in software engineering?	Software Evolution	Software Verification	Software Testing & Validation	Software designing	Software Verification
99	_____ originally proposed the spiral model.	Pressman	Royce	IBM	Barry Boehm	Barry Boehm
100	What does SDLC stands for?	System Design Life Cycle	Software Design Life Cycle	Software Development Life Cycle	System Development Life cycle	Software Development Life Cycle
101	What is the primary purpose of verification in software testing?	To ensure the software meets user requirements.	To check if the software is built according to specifications.	To identify defects in the software.	To validate user experience.	To check if the software is built according to specifications.
102	Which of the following is a key difference between verification and validation?	Verification is done before validation.	Validation is concerned with requirements; verification is not.	Verification involves actual execution of the software.	Both A and B.	Both A and B.
103	What does validation ensure in the context of software testing?	The software has no bugs.	The software meets the business needs and requirements.	The software complies with coding standards.	The software runs on the intended platforms.	The software meets the business needs and requirements.
104	What is the main goal of software inspections?	To find defects after testing.	To ensure code quality before implementation.	To document the development process.	To conduct performance testing.	To ensure code quality before implementation.
105	Which of the following is a benefit of conducting code reviews?	Decreased project cost.	Improved code readability and maintainability.	Increased number of defects.	All of the above.	Improved code readability and maintainability.
106	Which of the following best describes a formal review process?	Casual discussion among team members.	Structured meeting with defined roles and outcomes.	Ad hoc checking of code.	Peer testing without documentation.	Structured meeting with defined roles and outcomes.
107	What is a procedural unit in software design?	A block of code that performs a specific task.	A comprehensive documentation of software functions.	An individual software module.	A compilation of all software requirements.	A block of code that performs a specific task.

108	Which of the following is true about procedural units?	They can only be used in object-oriented programming.	They are always larger than 100 lines of code.	They promote modularity and reusability in software.	They cannot interact with other units.	They promote modularity and reusability in software.
109	What does size measure refer to in software metrics?	The performance of the software.	The amount of resources consumed by the software.	The quantifiable attributes of the software code.	The time taken to develop software.	The quantifiable attributes of the software code.
110	Which of the following is a commonly used size metric?	Lines of Code (LOC)	Function Points	Use Case Points	All of the above	All of the above
111	What is the primary purpose of complexity metrics?	To assess the cost of software development.	To measure the difficulty of understanding and maintaining software.	To evaluate user satisfaction.	To predict software performance.	To measure the difficulty of understanding and maintaining software.
112	Cyclomatic complexity is used to measure what aspect of software?	Memory usage	Control flow complexity	Code readability	User interface complexity	Control flow complexity
113	What does the Halstead measure evaluate?	The execution time of software.	The size and complexity of the software code.	The performance of software modules.	The user experience of the software.	The size and complexity of the software code.
114	Which of the following metrics is part of the Halstead measure?	Total number of operators	Cyclomatic complexity	Code coverage	Lines of code	Total number of operators
115	Cyclomatic complexity is calculated based on which of the following?	The number of inputs to a function.	The number of paths through a program.	The total number of lines of code.	The number of comments in the code.	The number of paths through a program.
116	What is the primary goal of unit testing?	To test the software as a whole.	To validate the interaction between components.	To verify the correctness of individual components.	To ensure user requirements are met.	To verify the correctness of individual components.
117	Which of the following tools is commonly used for unit testing?	Selenium	JUnit	JIRA	Git	JUnit
118	Which of the following best describes a unit test?	A test of the entire application.	A test of a specific function or method.	A test of user interfaces.	A performance test on the software.	A test of a specific function or method.
119	In which phase of software development is unit testing performed?	After deployment	During integration	During implementation	During requirements analysis	During implementation
120	What is a common advantage of unit testing?	It eliminates the need for system testing.	It allows for early detection of bugs.	It requires extensive documentation.	It increases development time significantly.	It allows for early detection of bugs.
121	What type of testing focuses on user requirements?	Unit Testing	Integration Testing	Acceptance Testing	System Testing	Acceptance Testing
122	Which of the following is NOT a verification activity?	Inspections	Walkthroughs	Testing	Reviews	Testing
123	Which type of metric helps in assessing maintainability?	Halstead Measure	Performance Metric	Security Metric	User Satisfaction Metric	Halstead Measure
124	What is the result of high cyclomatic complexity?	Easier to maintain code	Higher risk of bugs	Faster execution	Improved user experience	Higher risk of bugs
125	Which approach is best for unit testing?	Testing only at the end of development	Testing individual components as they are developed	Testing after deployment	Not testing at all	Testing individual components as they are developed

126	What is a common practice for effective unit testing?	Write tests after coding	Use mock objects for dependencies	Test only the happy paths	Avoid testing edge cases	Use mock objects for dependencies
127	Which of the following is a metric to assess code quality?	Test coverage percentage	Lines of code	Number of users	Time to market	Test coverage percentage
128	Which of the following statements is true about verification in the software development lifecycle?	Ensuring the product is built correctly	Ensuring the right product is built	User acceptance	Performance testing	Ensuring the product is built correctly
129	What type of testing ensures that individual components function correctly?	System Testing	Integration Testing	Unit Testing	Acceptance Testing	Unit Testing
130	Which technique is typically used during code inspections?	Automated testing	Static analysis	Performance testing	Dynamic analysis	Static analysis
131	Which of the following statements is true about unit tests?	They only test user interfaces.	They should be run before the code is deployed.	They are not essential for agile development.	They should only cover positive scenarios.	They should be run before the code is deployed.
132	What is the primary focus of quality assurance?	Finding defects in products	Preventing defects in processes	Ensuring customer satisfaction	Measuring product quality	Preventing defects in processes
133	Which of the following is a key activity in quality assurance?	Product testing	Process auditing	Defect tracking	Customer feedback	Process auditing
134	What does a quality assurance plan typically include?	Only testing methods	Project timeline	Standards and procedures	Marketing strategies	Standards and procedures
135	Which of the following is NOT a quality assurance activity?	Training personnel	Process definition	Product inspection	Process monitoring	Product inspection
136	What is the primary goal of quality control?	To develop processes	To ensure products meet specifications	To train employees	To improve production efficiency	To ensure products meet specifications
137	Which of the following tools is commonly used in quality control?	Flowcharts	Statistical process control (SPC)	Risk analysis	Cost-benefit analysis	Statistical process control (SPC)
138	What does quality control primarily involve?	Planning and documentation	Inspection and testing	Process improvement	Training and development	Inspection and testing
139	Which of the following best describes the concept of 'control limits' in quality control?	The maximum cost of production	The range of acceptable variation in a process	The minimum quality standard	The target market for a product	The range of acceptable variation in a process
140	Which of the following statements best describes the difference between QA and QC?	QA is process-oriented; QC is product-oriented.	QA focuses on documentation; QC focuses on training.	QA is done before testing; QC is done after testing.	Both A and C.	Both A and C.
141	Which of the following is a characteristic of quality assurance?	Reactive approach	Proactive approach	Focus on finished products	Inspection-oriented	Proactive approach
142	Which of the following is a characteristic of quality control?	It involves auditing processes.	It focuses on customer satisfaction.	It is preventive in nature.	It includes product testing.	It includes product testing.
143	Which activity would be considered part of quality assurance?	Product inspection	Process design review	Defect reporting	Corrective action	Process design review

144	What does CMM stand for?	Capability Maturity Model	Comprehensive Management Method	Controlled Management Model	Critical Maturity Model	Capability Maturity Model
145	Which of the following is NOT a maturity level in CMM?	Initial	Managed	Defined	Evaluated	Evaluated
146	What is the purpose of the CMM?	To establish quality standards	To assess and improve processes	To define project management techniques	To evaluate customer satisfaction	To assess and improve processes
147	At which maturity level are processes considered unpredictable and reactive?	Level 1: Initial	Level 2: Managed	Level 3: Defined	Level 4: Quantitatively Managed	Level 1: Initial
148	What is the primary focus of ISO 9001?	Environmental management	Quality management systems	Health and safety standards	Risk management	Quality management systems
149	Which of the following is a requirement of ISO 9001?	Continuous improvement	Annual financial audits	Employee training every month	Customer satisfaction surveys	Continuous improvement
150	What is the benefit of obtaining ISO 9001 certification?	Increased production costs	Enhanced customer satisfaction	Decreased employee involvement	Reduced product quality	Enhanced customer satisfaction
151	Which of the following is an essential principle of ISO 9001?	Focusing solely on the product	Involving top management in quality processes	Isolating quality activities from other operations	Avoiding documentation	Involving top management in quality processes
152	What is the primary goal of software reuse?	Increase software defects	Reduce development time and cost	Eliminate the need for documentation	Create complex systems	Reduce development time and cost
153	Which of the following is a benefit of software reuse?	Increased risk of software obsolescence	Lower development costs	Higher maintenance effort	Increased time to market	Lower development costs
154	What is a common approach to achieve software reuse?	Creating new software from scratch	Using design patterns	Ignoring existing code	Avoiding documentation	Using design patterns
155	Which of the following best describes a software component?	A complete software application	A reusable piece of software	A set of development tools	A user interface	A reusable piece of software
156	What is the main purpose of software reengineering?	To develop new software applications	To improve existing software systems	To eliminate software documentation	To increase software complexity	To improve existing software systems
157	Which of the following is NOT a phase of software reengineering?	Reverse engineering	Forward engineering	Code generation	Restructuring	Code generation
158	What is a common technique used in reengineering?	Code refactoring	Unit testing	Software deployment	User acceptance testing	Code refactoring
159	What does reverse engineering involve?	Developing new software	Analyzing existing software to understand its components	Testing software functionality	Writing user documentation	Analyzing existing software to understand its components
160	What is forward engineering?	Modifying existing software	Converting design specifications into code	Analyzing code to derive design	Documenting software	Converting design specifications into code
161	Which of the following describes a primary goal of reverse engineering?	To create user documentation	To understand system architecture	To design new algorithms	To test software	To understand system architecture

162	Which tool is commonly used in reverse engineering?	UML diagrams	Code analyzers	Integrated Development Environments (IDEs)	Version control systems	Code analyzers
163	What is a benefit of forward engineering?	Improved understanding of legacy systems	Enhanced code maintainability	Streamlined development processes	Increased software complexity	Streamlined development processes
164	What is a dependable system?	A system that is always available	A system that is secure and performs reliably	A system that is easy to use	A system with low operational costs	A system that is secure and performs reliably
165	Which of the following qualities is NOT typically associated with dependable systems?	Availability	Performance	Integrity	Inflexibility	Inflexibility
166	What is the main focus of dependable computing?	Cost reduction	System availability	User satisfaction	Software complexity	System availability
167	Which aspect is NOT a concern for dependable systems?	Security	Reliability	Usability	Development speed	Development speed
168	What does reliability engineering primarily focus on?	Ensuring that a system is secure	Ensuring that a system performs its intended function under specified conditions	Improving user interfaces	Reducing development time	Ensuring that a system performs its intended function under specified conditions
169	Which of the following is a key metric in reliability engineering?	Mean Time Between Failures (MTBF)	Code coverage	Lines of code	User satisfaction score	Mean Time Between Failures (MTBF)
170	What is the goal of conducting reliability tests?	To validate performance	To ensure compliance with standards	To identify potential defects	To assess usability	To identify potential defects
171	Which of the following methods is commonly used in reliability engineering?	Stress testing	Performance testing	Unit testing	Regression testing	Stress testing
172	What is the primary focus of safety engineering?	Preventing software defects	Ensuring the safety of users and systems	Improving software performance	Enhancing user experience	Ensuring the safety of users and systems
173	Which of the following is a common technique used in safety engineering?	Failure Mode and Effects Analysis (FMEA)	Code reviews	Unit testing	Integration testing	Failure Mode and Effects Analysis (FMEA)
174	What is a critical aspect of safety-critical systems?	They require extensive documentation.	They must meet stringent reliability requirements.	They should be user-friendly.	They must be developed rapidly.	They must meet stringent reliability requirements.
175	What does the term "safety integrity level" (SIL) refer to?	The level of code quality	The degree of system reliability	The level of assurance that a system operates safely	The time required for system updates	The level of assurance that a system operates safely
176	What is the main focus of security engineering?	Enhancing user interfaces	Protecting systems from unauthorized access	Improving system performance	Increasing software complexity	Protecting systems from unauthorized access
177	Which of the following is NOT a goal of security engineering?	Confidentiality	Availability	Maintainability	Integrity	Maintainability
178	What is a common security engineering practice?	Code obfuscation	Performance tuning	Usability testing	System integration	Code obfuscation
179	What does "defense in depth" refer to in security engineering?	Relying on a single security measure	Implementing multiple layers of security controls	Focusing solely on perimeter security	Ensuring fast response times	Implementing multiple layers of security controls

180	What is the primary goal of resilience engineering?	Ensuring systems can recover from failures	Improving user experience	Increasing software complexity	Reducing development costs	Ensuring systems can recover from failures
181	Which of the following is a key principle of resilience engineering?	Predictability	Flexibility	Simplicity	Rigidity	Flexibility
182	What does a resilient system typically have?	High performance under normal conditions	The ability to adapt to changes and recover from disruptions	Minimal documentation	Simplified architecture	The ability to adapt to changes and recover from disruptions
183	Which approach is commonly used to enhance system resilience?	Adding redundancy	Increasing system complexity	Reducing system resources	Simplifying the user interface	Adding redundancy
184	What is a common challenge in software reuse?	High initial development cost	Compatibility with legacy systems	Reduced software quality	Increased documentation requirements	Compatibility with legacy systems
185	What is the relationship between reengineering and maintenance?	Reengineering is a subset of maintenance.	Reengineering and maintenance are the same.	Maintenance focuses on new development, while reengineering focuses on existing systems.	Reengineering is unrelated to maintenance.	Reengineering is a subset of maintenance.
186	Which of the following is an example of forward engineering?	Analyzing existing software to create documentation	Developing a new application from a set of requirements	Updating legacy systems	Extracting information from a database	Developing a new application from a set of requirements
187	What is a key difference between reliability and safety engineering?	Reliability focuses on user needs; safety focuses on system performance.	Reliability is about system availability; safety is about preventing accidents.	Reliability is only concerned with software; safety applies to hardware.	There is no difference.	Reliability is about system availability; safety is about preventing accidents.
188	What role does usability play in dependable systems?	It is not a concern.	It enhances user interaction and satisfaction.	It reduces development costs.	It simplifies the architecture.	It enhances user interaction and satisfaction.
189	What is the role of a hazard analysis in safety engineering?	To identify potential risks and develop mitigation strategies	To improve code quality	To assess user experience	To design new algorithms	To identify potential risks and develop mitigation strategies
190	What is the focus of resilience engineering in system design?	Minimizing costs	Enhancing performance	Building systems that can withstand and recover from unexpected disruptions	Simplifying development processes	Building systems that can withstand and recover from unexpected disruptions
191	Which of the following best describes a resilient system?	One that is inflexible and rigid	One that performs consistently well under all conditions	One that can adapt to changes and recover from disruptions	One that requires constant monitoring	One that can adapt to changes and recover from disruptions
192	What is the primary purpose of a Software Requirements Specification (SRS)?	To outline the development process	To detail the system's architecture	To specify what the software should do	To define the coding standards	To specify what the software should do
193	Which of the following is NOT typically included in an SRS document?	Functional requirements	Non-functional requirements	Design specifications	User interface requirements	Design specifications
194	Who is primarily responsible for reviewing the SRS?	Project managers	Software developers	Stakeholders and end-users	Quality assurance team	Stakeholders and end-users
195	What is a key benefit of having a well-defined SRS?	It increases the development time	It decreases the need for documentation	It helps avoid misunderstandings between stakeholders	It eliminates the need for testing	It helps avoid misunderstandings between stakeholders
196	In which phase of the software development lifecycle is the SRS typically created?	Implementation	Requirements analysis	Testing	Deployment	Requirements analysis
197	What does a Gantt chart primarily illustrate?	The organizational structure of a project	The sequence of tasks and their durations	The budget allocated for a project	The quality standards for a product	The sequence of tasks and their durations

198	Which of the following elements is typically included in a Gantt chart?	Task dependencies	Resource allocation	Project milestones	All of the above	All of the above
199	What is a key benefit of using a Gantt chart in project management?	It eliminates the need for communication among team members	It provides a visual representation of project timelines and task progress	It guarantees project completion on time	It is the only tool needed for project management	It provides a visual representation of project timelines and task progress
200	In a Gantt chart, what does a task bar represent?	The resources assigned to a task	The duration of a task	The cost of a task	The priority of a task	The duration of a task
201	How can a Gantt chart be useful for identifying potential project delays?	By showing the tasks that are completed	By highlighting tasks that overlap or have dependencies	By eliminating the need for task assignments	By focusing solely on the budget	By highlighting tasks that overlap or have dependencies